

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE																											
End Semester Examination – Summer Supplementary 2022																											
Course: B. Tech.	Branch: Computer	Semester: IV																									
Subject Code & Name: BTCOC305 Digital Electronics & Microprocessor																											
Max Marks: 60	Date: - -2023	Duration: 3 Hr.																									
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected to answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.																											
	(Level /CO)	Marks																									
Q. 1	Solve any two of the following:																										
A)	Write a short note on Binary Number System. Describe Binary-to-Decimal Conversion.	(L1/ CO1) 6 M																									
B)	Explain with an example NAND, NOR, EX-OR gate.	(L1/ CO1) 6 M																									
C)	If the waveforms of given figure are applied at the inputs of a 2-input OR gate, and applied at the input of a 2-input AND gate determine the output waveform. 	(L2/ CO2) 6 M																									
Q. 2	Solve any two of the following.																										
A)	Explain in detail Karnaugh Map Representation of logic functions. Prepare the truth table for K-map of following: <table><tr><td>CD/AB</td><td>00</td><td>01</td><td>11</td><td>10</td></tr><tr><td>00</td><td>1</td><td></td><td>1</td><td></td></tr><tr><td>01</td><td></td><td>1</td><td></td><td>1</td></tr><tr><td>11</td><td></td><td>1</td><td>1</td><td></td></tr><tr><td>10</td><td></td><td></td><td>1</td><td></td></tr></table>	CD/AB	00	01	11	10	00	1		1		01		1		1	11		1	1		10			1		(L2/ CO2) 6 M
CD/AB	00	01	11	10																							
00	1		1																								
01		1		1																							
11		1	1																								
10			1																								
B)	Simplify the K–map of the following using POS method: $f(A, B, C, D) = \Sigma m(4, 6, 10, 12, 13, 15)$	(L2/ CO2) 6 M																									
C)	Explain in detail Multiplexer and De-Multiplexer.	(L2/ CO2) 6 M																									
Q. 3	Solve any two of the following.																										

A)	Convert S-R Flip-Flop to a J-K Flip-Flop.	(L3/ CO3)	6 M
B)	Write any two application of Flip-Flops.	(L3/ CO3)	6 M
C)	Write a note on asynchronous sequential counter.	(L3/ CO3)	6 M
Q. 4	Solve all two of the following.		
A)	Draw and explain the architecture of 8086 microprocessor.	(L2/ CO4)	6 M
B)	Difference between 8085, 8086, & 80386 microprocessor.	(L2/ CO4)	6 M
Q. 5	Solve any Two of the following.		
A)	Describe Addressing modes of 8086.	(L3/ CO5)	6 M
B)	Difference between Compiler and Assembler.	(L2/ CO5)	6 M
	*** End ***		